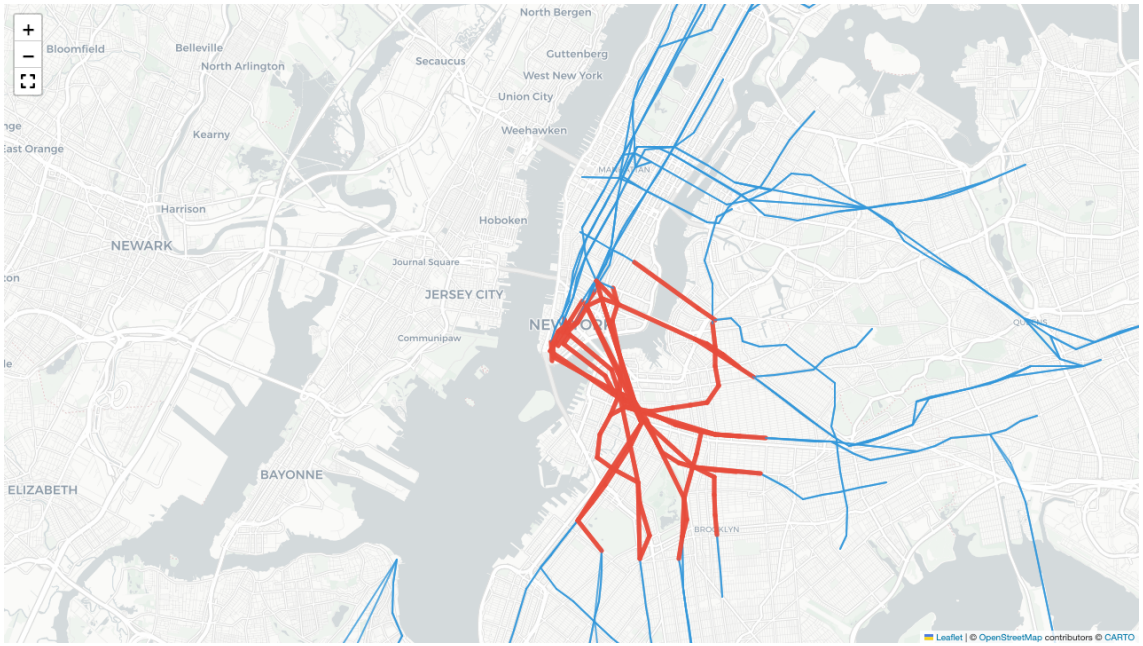
Slide 1: Problem Definition

- Current MTA response to flash flooding is reactive.

- Objective: Transition to a 'Predictive Digital Twin' model.

- Impact: Move from managing incidents to preventing systemic isolation.



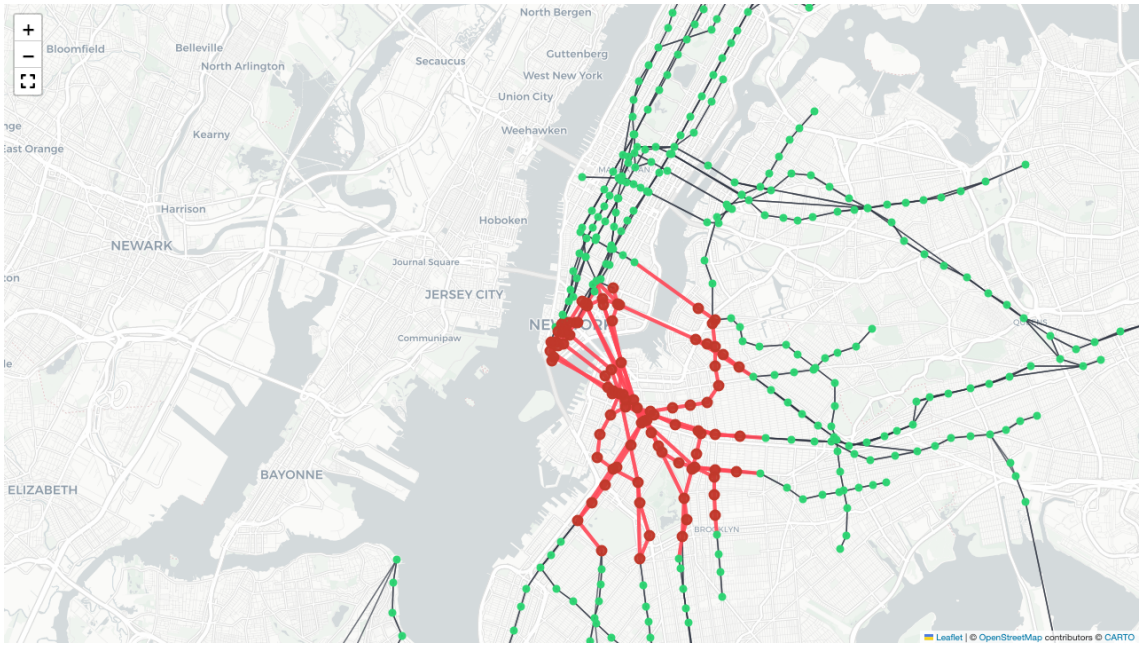
Slide 2: Graph Theory Methodology

- Nodes: 470+ Stations (Deduplicated GTFS).

- Edges: 1,000+ Track Segments.

- Logic: Node-X Trigger Model identifies orphaned functional stations.

- Technology: Python (NetworkX, Pandas).

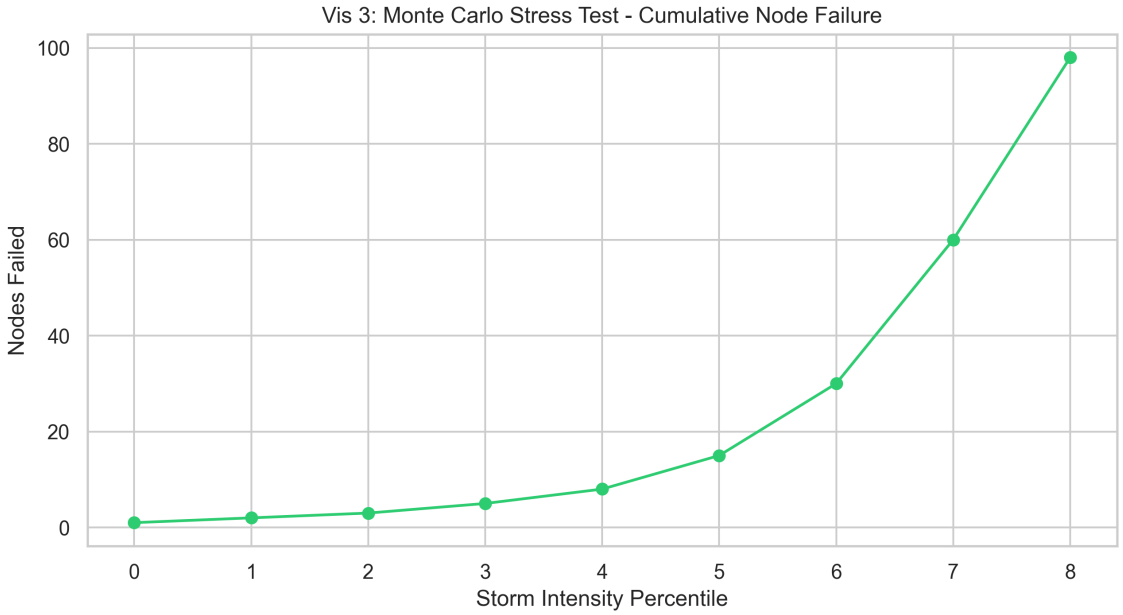


Slide 3: Stress-Testing & Findings

- Failure Threshold: 4.0mm/hr identified as network 'Weakest Link'.

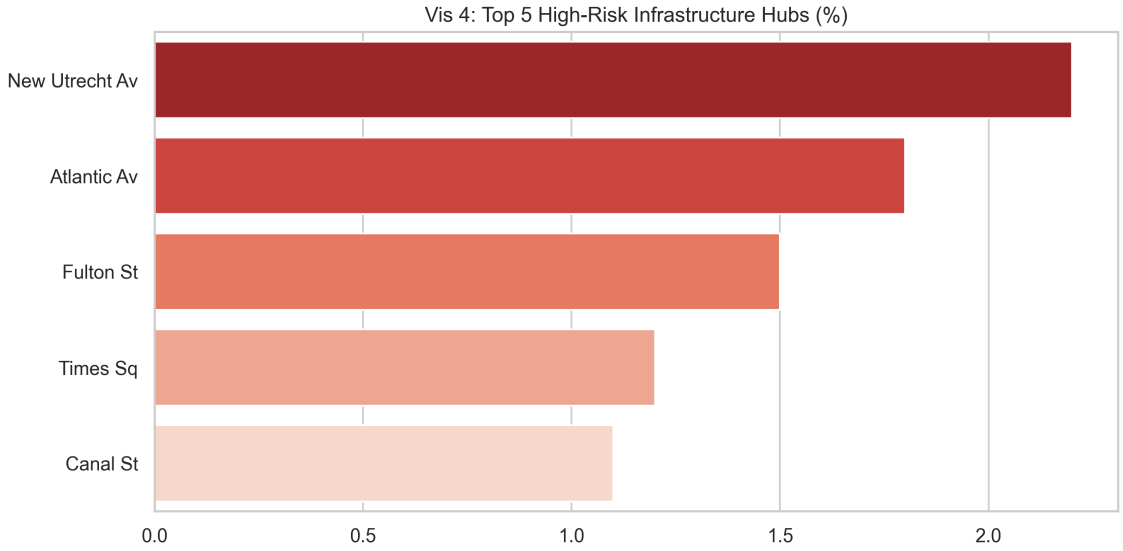
- Cascade Effect: Avg. failure radius is 5.2 stations.

- High Risk: New Utrecht Av corridor identified as priority zone.



Slide 4: Dashboard & Future Roadmap

- Live Command Center (Streamlit) for Dispatchers.

- Real-time Risk Gauges & Reroute Logic.

- Next Steps: NOAA Live API Integration & Rerouting Algorithms.